

Ninth International Olympiad, 1967

1967/1

Let $ABCD$ be a parallelogram with side lengths $AB = a$, $AD = 1$, and with $\angle BAD = \alpha$. If $\triangle ABD$ is acute, prove that the four circles of radius 1 with centers A, B, C, D cover the parallelogram if and only if

$$a \leq \cos \alpha + \sqrt{3} \sin \alpha.$$

1967/2

Prove that if one and only one edge of a tetrahedron is greater than 1, then its volume is $\leq \frac{1}{8}$.

1967/3

Let k, m, n be natural numbers such that $m + k + 1$ is a prime greater than $n + 1$. Let $c_s = s(s + 1)$. Prove that the product

$$(c_{m+1} - c_k)(c_{m+2} - c_k) \cdots (c_{m+n} - c_k)$$

is divisible by the product $c_1 c_2 \cdots c_n$.

1967/4

Let $A_0 B_0 C_0$ and $A_1 B_1 C_1$ be any two acute-angled triangles. Consider all triangles ABC that are similar to $\triangle A_1 B_1 C_1$ (so that vertices A_1, B_1, C_1 correspond to A, B, C , respectively) and circumscribed about $\triangle A_0 B_0 C_0$ (where A_0 lies on BC , B_0 on CA , and C_0 on AB). Of all such possible triangles, determine the one with maximum area, and construct it.

1967/5

Consider the sequence $\{c_n\}$, where

$$c_1 = a_1 + a_2 + \cdots + a_8,$$

$$c_2 = a_1^2 + a_2^2 + \cdots + a_8^2,$$

$$\vdots$$

$$c_n = a_1^n + a_2^n + \cdots + a_8^n,$$

in which $a_1, a_2, \dots, a_8$ are real numbers not all equal to zero. Suppose that an infinite number of terms of the sequence $\{c_n\}$ are equal to zero. Find all natural numbers n for which $c_n = 0$.